

Harnessing Trigger Factors for Headache Management

Timothy T. Houle, PhD



MASSACHUSETTS
GENERAL HOSPITAL

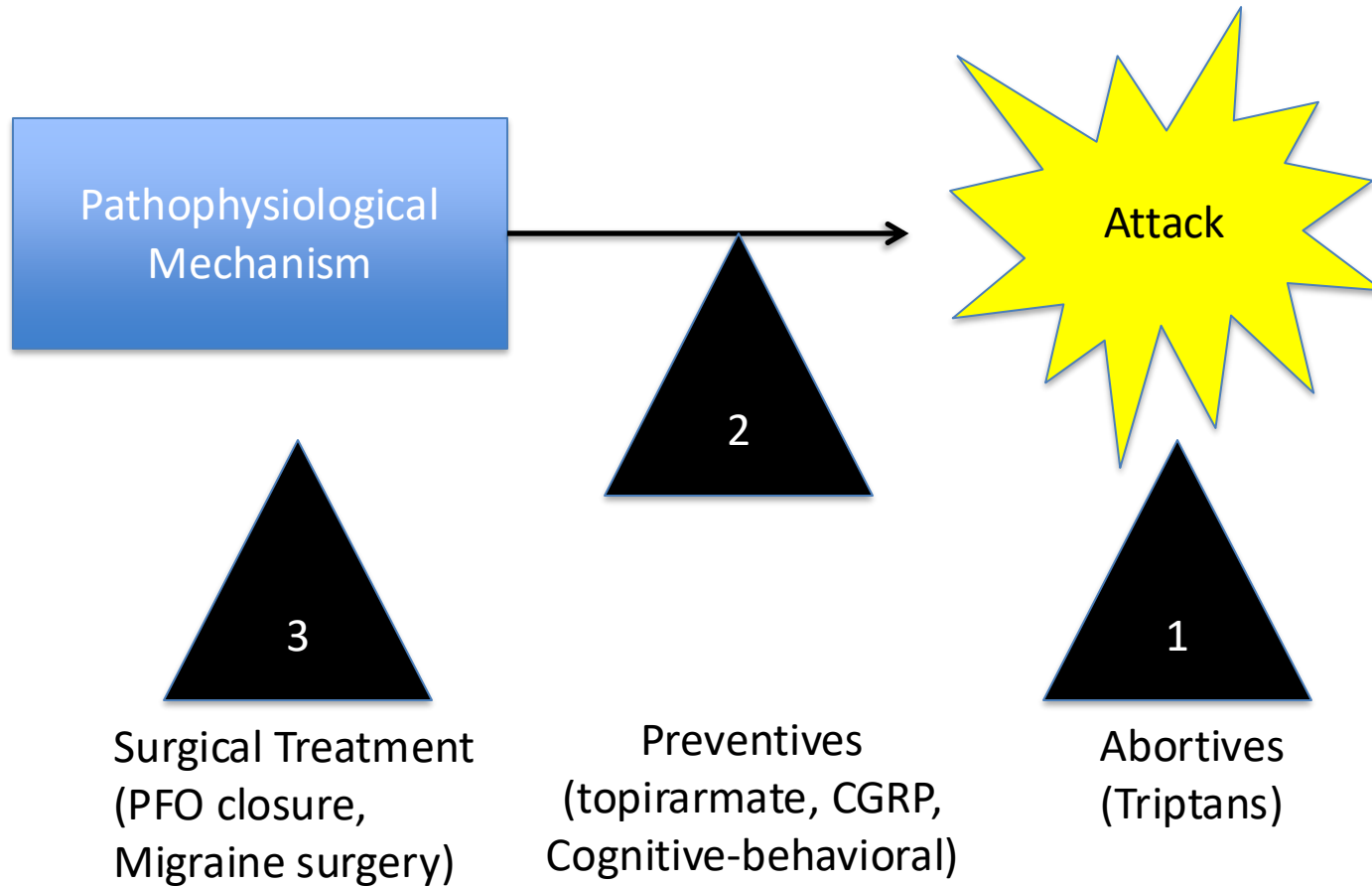


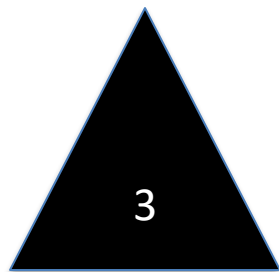
HARVARD
MEDICAL SCHOOL

Conflicts of Interest

- No financial relationships with industry
- Statistical Editor/Consultant
 - Anesthesiology
 - Headache
- Funding: National Institute of Neurological Disorders and Stroke (NINDS)
 - RO1NS06525701

Current Treatment Paradigms





Surgical Treatment

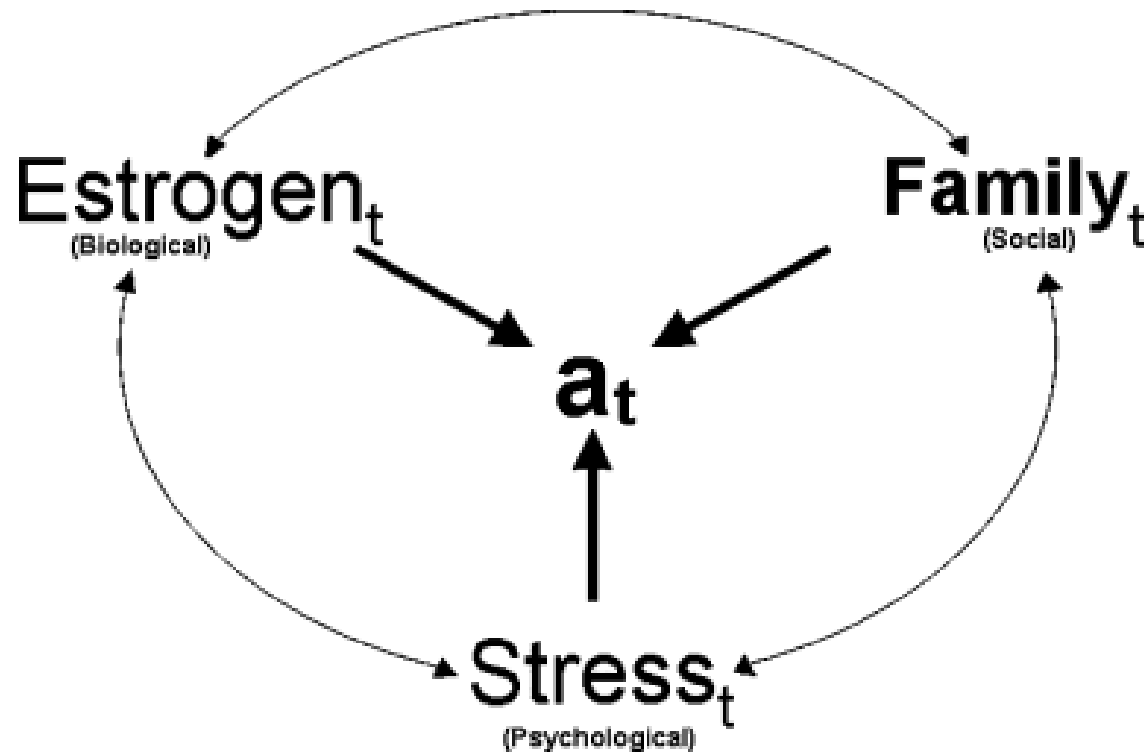
Delivery	Daily Risk Profile	Comment
On or After attack occurs	Very High	Suffering risk of progression
Daily or Monthly	> Moderate (indiscriminate)	Poor adherence
Once or infrequently	High (regression to mean)	(Unknown)

Preemptory Treatment*

	Delivery	Daily Risk Profile	Comment
Abortive	On or After attack occurs	Very High	Suffering risk of progression
Preventive	Daily or Monthly	> Moderate (indiscriminate)	Poor adherence (Expensive)
Surgical	Once or infrequently	High (regression to mean)	(Unknown)
Preemptive	During high risk period(s)	Moderate to high	Requires forecasting

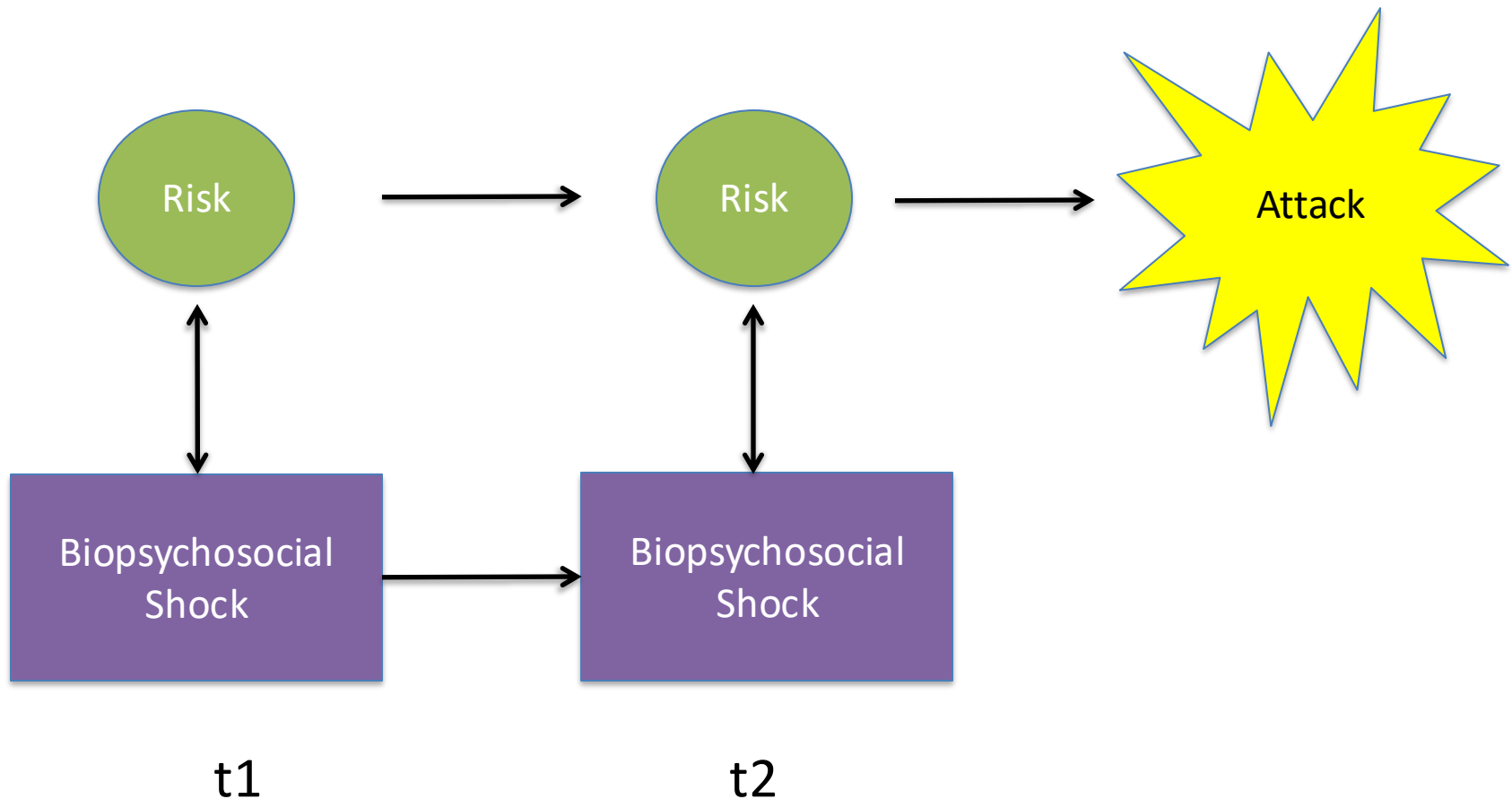
*Giffin NJ, Ruggiero L, Lipton RB, et al. Premonitory symptoms in migraine: An electronic diary study. Neurology. 2003;60:935-940

A Forecasting Treatment Paradigm: Biopsychosocial Process Model



Houle TT, Remble TA, Houle TA. The examination of headache activity using time series research designs. *Headache J Head Face Pain*. 2005;45:438-444.

A Forecasting Treatment Paradigm

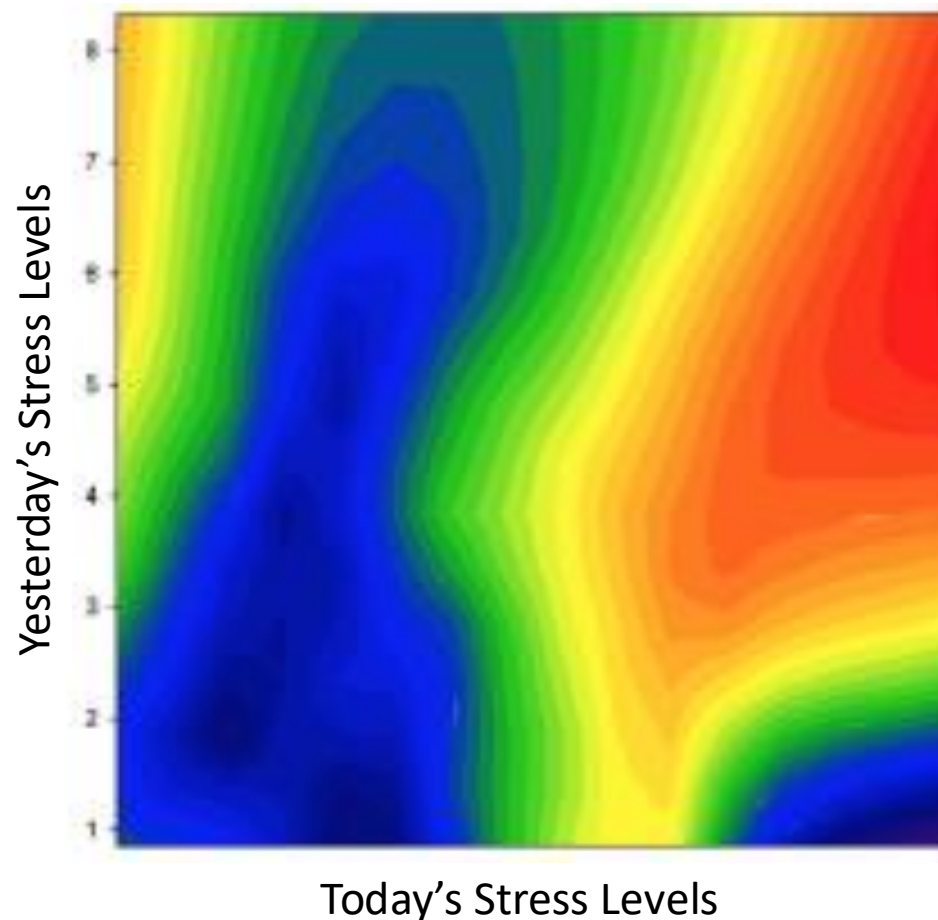


Stress & Headache

- Long history with mixed findings
 - Positive association in diary studies
 - Lack of association in biomarkers (cortisol)
 - Heterogeneity in self-report
- Types of stress (Selye)
 - Major life events
 - Daily hassles
- “Let-down” headache (¹⁻³)
 - High stress leads to increased headache AFTER stress subsides

Stress and Sleep Duration Predict Headache Severity in Chronic Headache Sufferers

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Headache Prediction Study (HAPRED)

- Prospective longitudinal cohort study of episodic migraine sufferers Winston-Salem, NC region from September 2009 to May 2014.
- Participants were recruited from the general community, neurology practices, and primary care clinics using direct television and print (e.g., fliers) advertising.
- Inclusion criteria:
 - either gender
 - episodic migraine with or without aura (IHS-II 2.1 or 2.2)
 - > 2 headache attacks/month (4 and 14 headache days/month)

Headache Prediction Study (HAPRED)

- Exclusion criteria:
 - the presence of a secondary headache disorder (e.g., brain tumor)
 - a recent change in nature of headache symptoms over last 6 weeks,
 - not being able to read or speak English at a 6th grade level
 - having a psychiatric hospitalization within the previous 12 months
 - meeting criteria for, or currently being treated for a DSM-IV diagnosis of substance abuse (not including AA, NA, etc.)
 - pregnancy or planned pregnancy during the observation period
 - lack of stable residence or contact number
 - meeting IHS-II criteria for medication overuse (greater than two days/week of abortive medication use for headache)
- Headache diagnosis and other medical exclusions were confirmed by a Neurologist using the structured diagnostic interview

Daily Assessment of Stress & Headache

- Participants completed two electronic diary entries each day (i.e., AM and PM).



Daily Assessment

- Headache Activity
- Sleep
- Stress
- Mood
- Diet (caffeine, alcohol, missed meals)
- Common triggers (certain foods, menstrual cycle)

Daily Stress Inventory (DSI)

Brantley et al, 1987

- 58 item scale that allows a subject to identify stressful events (i.e. daily hassles) that they have experienced in the last 24 hours
- “Interrupted during task/activity”
- “Waited longer than you wanted”
- Occurred (yes or no, rated 1-7)
 - 1 “occurred, but not stressful”
 - 7 “caused me to panic”

Subscales

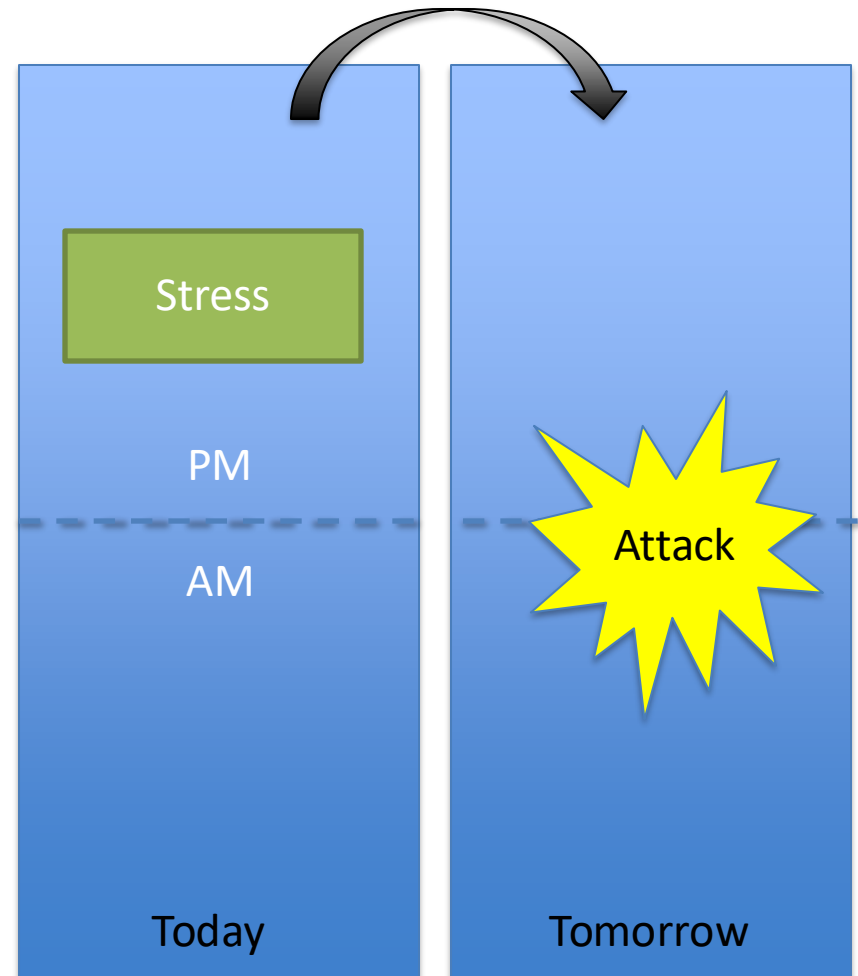
Frequency
(0 to 58)

Sum
(0 to 406)

Average
Intensity Rating
(0 to 7)

Primary Outcome

- Prediction (forecast) of a future headache attack based on current levels of stress and headache.
- The primary outcome is the presence/absence of any headache attack (head pain > 0 on a NRS) occurring over the 24 hour period recorded following a PM diary entry.



Results

- N = 95/100 (95%) participants completed the study
 - 5 drop-out (contributed negligible data)
- 4626 diary days were completed
 - Mean 49 days (range: 11 – 96)
- 431 (9.3%) embedded missing diary days

Participant Characteristics	N = 95
Age	40.3 (12.9)
Gender	
Female	86 (90.5%)
Male	9 (9.5%)
Race/ethnicity	
Caucasian	83 (87.4%)
African American	9 (9.5%)
Other	3 (3.1%)
Diagnosis	
Migraine without aura	79 (83.2%)
Migraine with aura	10 (10.5%)
Migraine with TTH	6 (6.3%)
MIDAS	10 [3.25, 18]
CES-D	9 [4, 17.5]
STAI	36 [32, 43.5]

Mean (SD), Frequency (%), Median [25th, 75th]

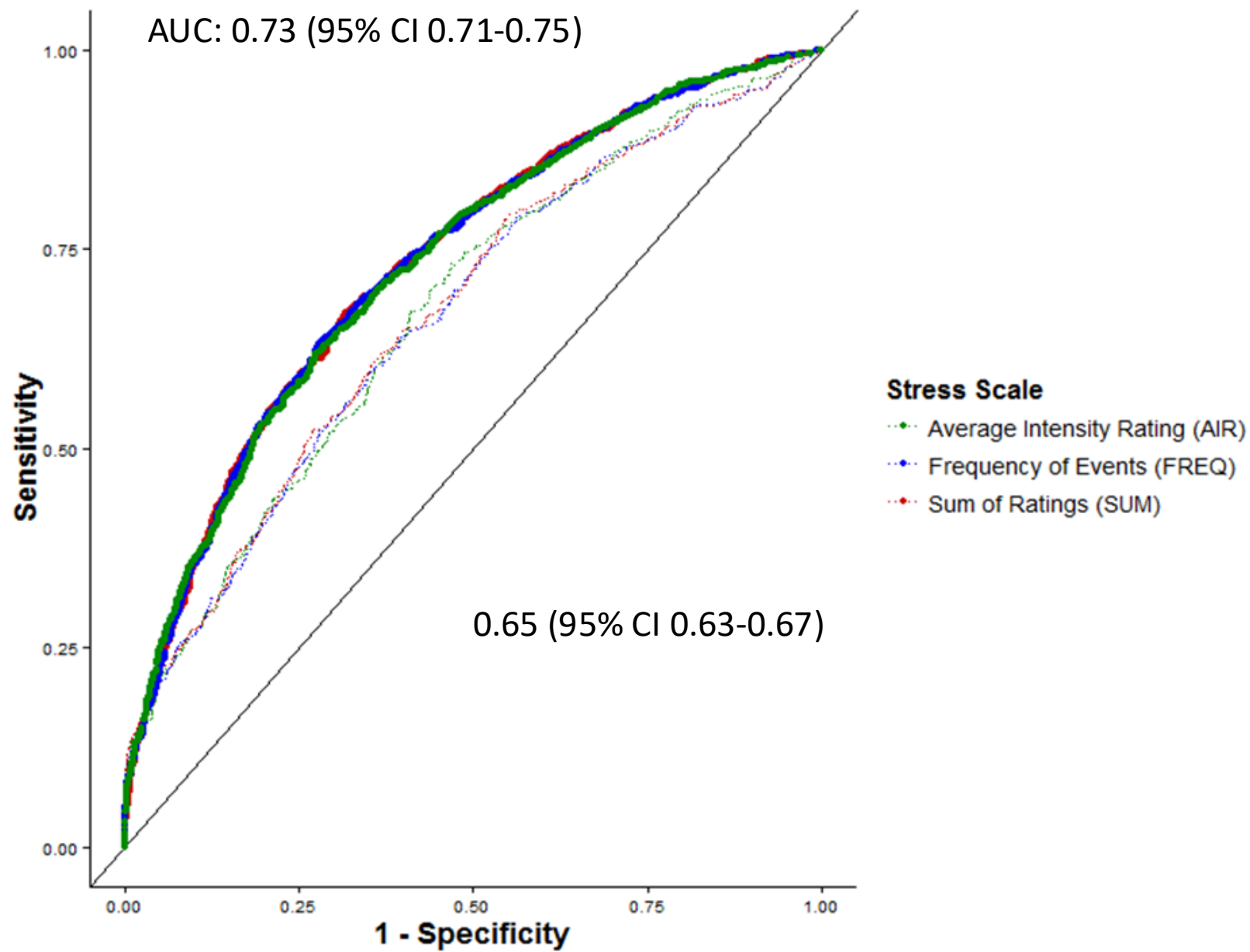
Day-level Characteristics:

	No Headache Tomorrow	Headache Tomorrow	P-value
N	2241	1437	
Stress			
FREQ	5 [2, 9]	6 [3, 11]	<0.001
SUM	11 [4, 21]	14 [7, 27]	<0.001
AIR	2 [1.29, 2.94]	2.4 [1.7, 3.0]	<0.001
Headache Today	765 (34.1%)	698 (48.6%)	<0.001

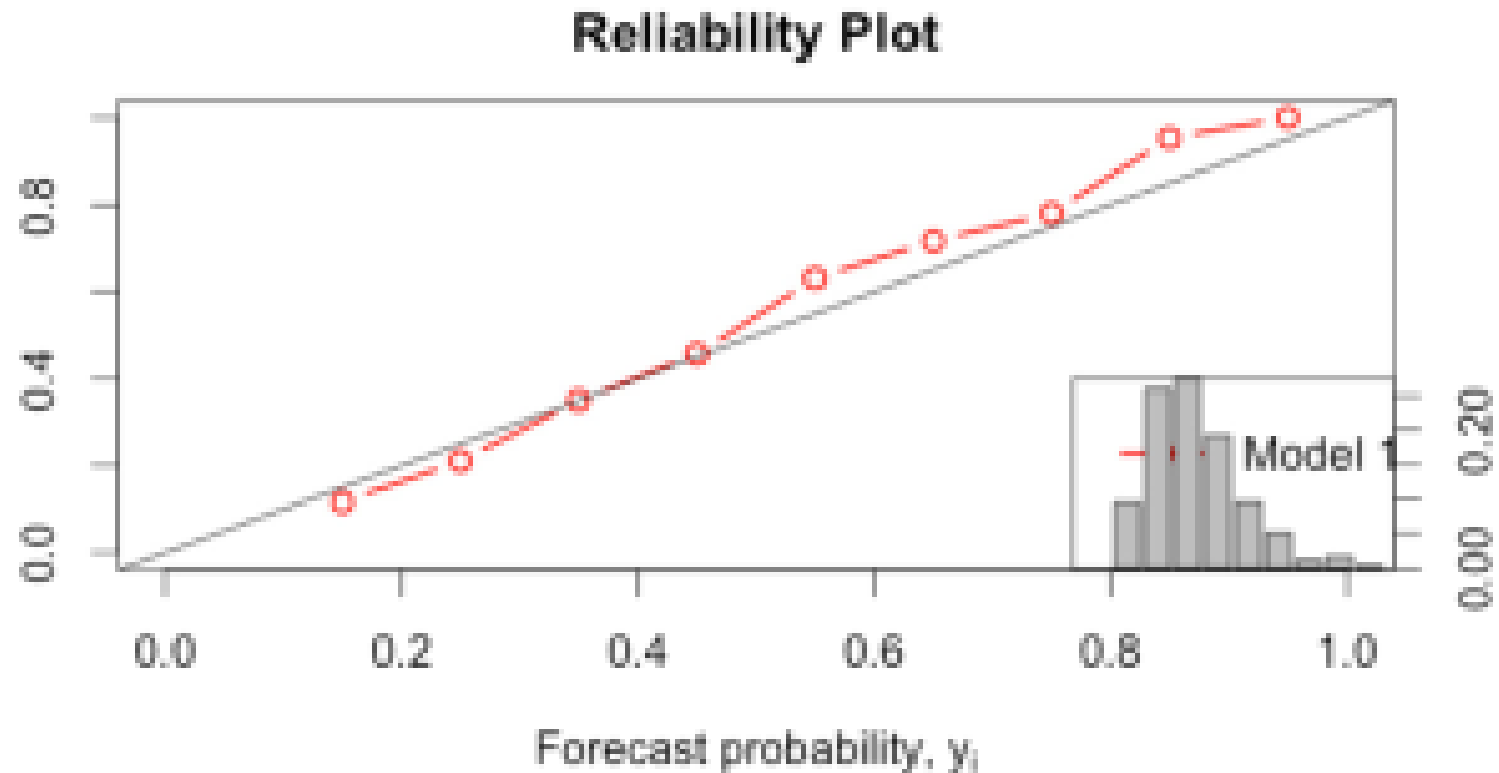
Headache occurred on: 1613/4195 (38.5%) days

Prediction Model

- Today's headache was associated with 45% increase in the odds of headache tomorrow
 - OR 1.45 (95%CI: 1.15 to 1.82), $p = 0.0018$
- Each point in today's stress (AIR) was associated with an 23% increase in the odds of headache tomorrow
 - OR 1.23 (95%CI: 1.12 to 1.34), $p = <0.0001$



Calibration & Resolution



Summary

- Perceived stress in the form of 'daily hassles' could be used to forecast headache risk
 - The predictive power was modest (AUC .65 to 0.73)
 - The calibration was good
 - The resolution was poor
- The model is not ready for clinical use, but this simple model represents a big first step to precise preemptive treatment
- The use of headache triggers to guide preemptive treatment has a vast potential to reduce suffering

Contributors

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